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Artificial intelligence activities and ethical approaches in leading listed companies in the European Union

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ABSTRACT

This study explores the information regarding Artificial Intelligence (AI) included by European listed companies in their annual and/or sustainability reports. The study mainly focuses on (1) the development and use of AI systems/projects reported by companies, (2) the extent to which companies disclose ethical principles or guidelines regarding AI and (3) the factors explaining these practices.

The study analyses the reports of 200 companies listed in the major indexes of Germany, Sweden, Finland, France, Spain, and Italy, both from qualitative and quantitative perspectives. All reports are analysed, using content analysis methodology, to identify expressions such as 'artificial intelligence', 'machine learning', 'deep learning', and 'big data', and then classified accordingly. The study's findings suggest a growing interest in the above-mentioned technologies, although 41.5% of companies do not report any activity in the field of AI. The adoption of ethical approaches to AI is at a very preliminary stage, and<5% of companies report on that issue. The quantitative analysis shows that larger companies, companies in the Technology and Telecommunications industries, and companies based in Southern countries are more likely to disclose information on AI activity. The majority of companies that develop ethical principles are listed in the Northern region and belong to the Technology and Telecommunications industries.

The study provides evidence of AI disclosure, a type of non-financial disclosure that has not been explored yet in the literature. Unlike existing studies, we propose a first definition of the topic and a taxonomy that can be used in further research on AI disclosure and can contribute to the development of KPIs in the field. Furthermore, this study provides a theoretical framework integrating some traditional theories, such as Voluntary disclosure theory, Signalling theory, and Legitimacy theory, specifically drawn to interpret AI disclosure practices, which can help with a further in-depth exploration of AI disclosure combining concurrent perspectives. The study's results may serve as a starting point for researchers and companies interested in the topic.

1. Introduction

Artificial Intelligence (AI) is considered a cornerstone of the economic and social development of the 21tst century (Annoni et al., 2018). It is applied in various fields for different purposes, from autonomous guide to medical diagnosis. In the years to come, the

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growing interest in AI is expected to lead to its widespread adoption, thus achieving a better and sustainable way of living and doing business, and satisfying societal needs (European Commission, 2018). In this context, various questions arise about the extent to which AI affects businesses, consumers, and the whole economy.

From the firm's perspective, the adoption of AI generates new benefits, possibly leading to a substantial increase in labour productivity, efficiency, and new revenue streams (Pwc, 2017).

At the same time, as other emerging technologies (Rotolo et al., 2015), the development, deployment, and use of AI systems imply several risks and ethical challenges due to their impact on human rights, privacy, employment, and other social issues (European Commission, 2019b). The application of AI technologies may lead to unfair results or discrimination as AI algorithms may be imperfect (Martin, 2019), lack transparency (Vakkuri et al., 2019b). In addition, the available data may be insufficient or contain distorted information (European Commission, 2019b).

In the last few years, the European Union (EU) and various European countries have been encouraging and regulating AI innovation. Furthermore, companies have begun disclosing information regarding their AI initiatives. However, in Europe, AI disclosure is voluntary; hence, whether to disclose and to what extent, and the type of disclosed information are almost entirely left to the discretion of companies. The lack of a shared vision or shared standards regarding the disclosure of AI activities leads to different AI disclosure practices of companies, depending on their perception.

The European Commission has begun discussing AI in the effort to solve key societal and human issues and establish the AI vision, strategy, and priorities of the EU (European Commission, 2018). In 2019, the European Commission published the 'Ethics Guidelines for Trustworthy Artificial Intelligence', which, inspired by fundamental rights and ethical principles, define seven key requirements that AI systems should meet to be trustworthy. The document sets general requirements to foster trustworthy AI in every field of human life, at every level of social organisation, and for every application. However, an additional effort may be needed for declining those requirements in specific contexts, such as business, industries, and related activities. The document requires fairness in terms of equal distribution of benefits and costs of AI systems, clear accountability of the AI systems use, and explicability in terms of transparency and disclosure of the AI system capabilities and limitations and ethics requirements.

Finally, considering the growing relevance of AI for sustainable development and the importance of disclosing AI information to the stakeholders, the aim of this research work is to explore information about AI activities and ethical principles concerning AI, which the European listed companies include in their annual and/or sustainability reports. Using the lenses of Voluntary Disclosure Theory (VDT), Signalling Theory (ST), and Legitimacy Theory (LT), this study provides an interpretative framework for corporate voluntary disclosure of information regarding AI development, deployment, and use. VDT, ST, and LT allow us to interpret the role of corporate AI information, both in competing in the capital markets to attract investors and aligning to society norms, beliefs, and values, to gain social acceptance from the stakeholders. However, the literature mainly focuses on the role of voluntary disclosure, signalling, and legitimacy in non-financial information, such as Corporate Social Responsibility (CSR), but largely ignores AI discourse.

Hence, this study addresses the following research questions:

RQ1. Are European listed companies disclosing information about AI?

RQ2. Are they reporting on AI activity?

RQ3. Are they disclosing AI ethical principles?

RQ4. What are the factors associated with reporting on AI activities or ethical approaches to AI?

The remainder of the paper is organised as follows. Section 2 presents the institutional context. Section 3 describes VDT, ST, and LT with a specific focus on non-financial and CSR disclosure and develops the theoretical framework. The methodology and results are presented in Sections 4 and 5, respectively. A discussion of the results and our concluding remarks are presented in Sections 6 and 7, respectively.

2. Institutional context: AI discourse in the European Union

Since 2018, the EU has contended that AI offers substantial opportunities to address societal challenges, such as sustainable development, gender balance, health enhancement, climate change, and the circular use of resources (European Commission, 2018). However, AI may also entail pitfalls and risks, such as limitations to human freedom, compromising human dignity, undermining democratic governments, and enhancing disparity. The approaches adopted by EU countries to manage AI development, deployment, and use will shape the future world (European Commission, 2018). In the EU vision, as per the Statement (2018) of the European Group on Ethics (EGE) in Science and New Technologies, AI should be seen as a tool operating in the service of humanity and public good to increase individual and collective well-being. Hence, AI should be managed with the scope to maximise the benefits and

Table 1
Trustworthy AI requirements according to the European Commission (2019).

Trustworthy requirements	Description
1 Human agency and oversight	Including fundamental rights, human agency and human oversight
2 Technical robustness and safety	Including resilience to attack and security, fall back plan and general safety, accuracy, reliability and reproducibility
3 Privacy and data governance	Including respect for privacy, quality and integrity of data, and access to data
4 Transparency	Including traceability, explainability and communication
5 Diversity, non-discrimination and fairness	Including the avoidance of unfair bias, accessibility and universal design, and stakeholder participation
6 Societal and environmental wellbeing	Including sustainability and environmental friendliness, social impact, society and democracy
7 Accountability	Including auditability, minimisation and reporting of negative impact, trade-offs and redress

prevent and minimise the risks that result from its adoption.

In 2019, the European Commission set up the High-Level Expert Group on Artificial Intelligence (AI-HLEG), an independent expert group, as part of its AI strategy. The group has drafted the Ethics Guidelines for Trustworthy Artificial Intelligence (European Commission, 2019b), inspired by the EGE document. Based on fundamental rights and ethical principles (respect for human autonomy, prevention of harm, fairness and explicability), the document identifies three essential components of trustworthy AI: lawfulness, ethicality, and robustness. It also establishes a framework for the last two components and provides general guidelines for operationalising ethical principles through seven key requirements (Table 1) that AI systems should meet, and key questions to assess trustworthiness. These requirements are of equal importance and should be implemented throughout the entire life-cycle of an AI system, that is, development, deployment and use.

The EU Framework proposes a comprehensive view of trustworthy AI with requirements based on a human-centric, socio-technical, life-cycle, and sustainable approach. The primacy of humans and human values should inspire AI systems in every field of human life and society, comprising their entire life-cycle. Trustworthiness also concerns AI regarded in a holistic and systemic perspective, that is, its technological elements and all relevant actors (developers, deployers, end-users, and society) and processes of socio-technical context of the AI system. Furthermore, establishing trustworthy AI systems requires a continuous process along their entire life-cycle from 'development (including research, design, data provision, and limited trials), deployment (including implementation) and use phase' (European Commission, 2019b, p. 36). Finally, AI systems must contribute to the Sustainable Development Goals (United Nations, 2015) and be managed 'enabling the flourishing of future generations to come' (European Commission, 2019b, p. 37).

In the document, companies are considered AI stakeholders (as end-users). They are required to adapt the general horizontal AI Trustworthiness framework to specific context as business sectors and AI-applications (European Commission, 2019b), and to be transparent and accountable for AI adoption, communicating 'in a clear and proactive manner, information to stakeholders about the AI system's capabilities and limitations, enabling realistic expectation setting, and about the manner in which the requirements are implemented' (European Commission, 2019b, p.18).

3. Literature review

3.1. AI and ethics

Researchers have long investigated the impact and potential of AI, and the ethical implications of its use in several fields (i.e., auditing, autonomous cars, social media, maritime sector, etc.) (Vakkuri and Abrahamsson, 2018; Munoko et al., 2020). Attention has been devoted to technological issues and theoretical questions (Vakkuri and Abrahamsson, 2018). Recently, academic discussion on AI is gaining momentum as its employment is gradually spreading to various human and business activities (Vakkuri and Abrahamsson, 2018). AI is incorporated in a growing number of devices, services, and machines. The current development of digital technologies, relatively low cost of computing power, availability of data (Big Data), and current generation of AI tools, which can learn from data (Munoko et al., 2020), among others, lead to an in-depth investigation of the implications of AI.

Scientific research considers the impact of AI on workers, companies, consumers, and society in a mixed way. On the one hand, it highlights its positive implications, such as an increase in productivity, better working conditions and safety, and the creation of new jobs (Wright and Schultz, 2018). It also underlines that AI positively affects larger firms, which invest enough financial resources in the automation of processes, and stresses that consumers can benefit from lower prices. On the other hand, AI is associated with the risk of economic stagnation, inequality, replacement of employees, and lack of satisfaction of basic human needs, among others. From this perspective, developed countries may benefit more than developing economies from the growing adoption of automation thanks to the reduction of labour costs and the decreased convenience of outsourcing (Wright and Schultz, 2018).

Due to the above-mentioned controversial role of AI and its nature as an emerging technology, characterised by 'radical novelty, relatively fast growth, coherence, prominent impact, uncertainty, and ambiguity' (Rotolo et al., 2015, p. 1839), many studies focus on its ethical implications. However, research concerning this topic is still at its infancy, mostly theoretical, and typically oriented to the discussion of the future influence of AI (Vakkuri and Abrahamsson, 2018). Researchers call for deeper investigation, empirical analysis, and real case studies. How various theoretical frameworks may be translated into 'practical implementation of ethics into AI systems' (Vakkuri and Abrahamsson, 2018, p. 5) is still unclear.

Despite the call for investigation of practical applications, some studies develop general theoretical frameworks of the ethical implications of AI starting from models built for emerging technologies. These studies focus on specific fields of application such as auditing (Munoko et al., 2020) or integrate different theories as stakeholder theory and social contracts theory (Wright and Schultz, 2018). Others analyse the ethical implications of AI through a systematic literature review, addressing accountability as well as specific components of AI as algorithms (Wieringa, 2020). Most studies focus on issues of the introductory phase of AI adoption (Allen et al., 2005; Stahl et al., 2010; Wright, 2011; Brey, 2012) or the developing phase (Palm and Hansson, 2006).

Little is known about the practical application of AI to various fields. Its ethical impact is not only related to the type of technology but also to the context in which it is used and how it is employed. Hence, the most recent AI ethical frameworks adopt a forecasting approach. This proactive strategy aims at preventing ethical issues by discussing and forecasting them with appropriate methodologies. It is a promising approach because AI implications are more evident in the long run (Munoko et al., 2020). Different methods are employed to identify technologies, artefacts, applications, and ethical issues, such as bibliometric analysis (Stahl et al., 2010; Munoko et al., 2020), experts surveys, roundtable discussions (Palm and Hansson, 2006; Wright, 2011), policy documents, and company studies (Brey, 2012).

As mentioned above, frameworks based on a forward-looking approach consider three relevant dimensions (technology, artefacts,

and application) for ethical implications (Brey, 2012), seen as pillars of AI governance and policies for managing and mitigating ethical risks (Munoko et al., 2020). In particular, the application level refers to the specific contexts and/or particular purposes and/or specific group of users in which AI is employed. Often, 'more specific ethical issues may emerge from the dynamic interplay between artefact and its contextual elements' (Brey, 2012, p. 311).

Finally, most existing ethical models of emerging technologies and AI propose a checklist of values or principles in line with the EU Guidelines. Such principles are used to identify ethical issues related to specific technologies and assess their likelihood and significance (Brey, 2012; Munoko et al., 2020).

3.2. Theoretical approaches to voluntary AI disclosure: An integrated framework

As underlined in the previous paragraphs, AI adoption has a pivotal role in future sustainable development. However, simultaneously, it has some social and ethical implications for firms, their stakeholders, and their reference context. Furthermore, in the near future, regulations will demand initiatives to assure transparency and trustworthiness in AI development, deployment, and use (European Commission, 2018). Therefore, disclosing voluntary information could be one of the most effective means that companies should adopt to satisfy those different needs.

According to the existing literature on firms' disclosure (Dye, 2001; Cho et al., 2015; van Zijl et al., 2017), VDT, ST, and LT are frequently used to explain non-financial reporting and voluntary disclosure. VDT and ST are considered economic-based theories (Lu and Wang, 2021); they view CSR as 'an addendum to conventional accounting activity' and elevate the financial community as the primary user of financial and non-financial information disclosed (Gray et al., 1995, p. 48). However, LT is regarded as a socio-political perspective that emphasises the role of 'information in organization-society dialogue' (Gray et al., 1995, p. 48). We use multiple theories capable of giving simultaneous lenses to the interpretation of voluntary AI disclosure of European listed companies and delivering an accurate understanding of the motivations of this disclosure. VDT, ST, and LT support our investigation and interpretation of voluntary AI disclosure to achieve a comprehensive view of the motivations subtended. The concomitant adoption of those views, in our opinion, may be able to capture the multifaceted and controversial role of AI in today's society and markets and can help effectively understand the behaviours of listed companies in their approach towards strategic and complex technology. Furthermore, the adoption of a multi-theoretical perspective answers a specific call for research in the field of non-financial disclosure (An et al., 2011).

Voluntary Disclosure Theory

VDT has its roots in the game theory and states that companies or managers voluntarily disclose information favourable to them and withhold unfavourable information (Dye, 2001). Studies investigate how companies use voluntary disclosure of information to influence and change investor behaviours or analyst forecasts (Zhou et al., 2017) and how managers and firms use their discretion to decide what information to disclose and what to withhold. VDT can interpret companies' and managers' behaviours, analysing motivations and incentives in disclosing information to the capital market (Shehata, 2014). In fact, the point of the dissemination of information or the degree of information quality provided to the market is related to the reduction of the cost of capital (Verrecchia, 2001), the risk, the cost in disclosing proprietary information (Verrecchia, 1983), and the reduction of information asymmetry (Shehata, 2014).

This theory has been initially applied to financial and accounting information and investors' behaviour. Subsequently, it has been expanded to non-financial information and communication and several types of stakeholders (Júnior et al., 2014). Studies assert that financial and non-financial information is interactively used by analysts to capture a reliable picture of the future performance of firms (Zhou et al., 2017).

According to this view, one of the most important means to disclose information are the reports published by companies, both mandatory or voluntary, financial and non-financial (Júnior et al., 2014). Managers exercise their discretion to identify the threshold level of disclosure and consequently decide what and when to disclose according to the investors' expectations (Verrecchia, 1983). Information disclosed through these reports is an input for the decision-making process of investors and the other agents in the financial market. The aim is to overcome the minimum level of information disclosed through mandatory information and reports and then reduce the information asymmetry between managers and investors (Júnior et al., 2014). Thus, the literature focuses mainly on the key attributes and contents of voluntary disclosure that influence investors' behaviour. Furthermore, theoretical studies investigate managers' decision-making processes in defining the amount, quality, and timing of information disclosure (Zhou et al., 2017).

Managers' discretion about information disclosure depends on several firm-specific factors such as size, performance, and corporate governance practices (Júnior et al., 2014, p. 182).

VDT is suited to provide a particular way of interpretation for the communication of information relating to AI initiatives towards the capital market and investors. According to VDT, companies are incentivised to disclose positive information about their development, deployment, and use of artifacts and applications based on AI technologies. AI voluntary disclosure can help investors grasp a further broadened estimate of companies' investments in smart technologies, overcoming information asymmetry. Elements such as the amount of information disclosed, the quality, and the timeliness can help investors forecast the future development of the company. Furthermore, information disclosed in the annual report regarding AI initiatives is the output of top manager's judgement of what information about AI is relevant to compete in the capital market, to obtain positive effects in terms of economic impact and investor attraction.

Signalling theory

ST has its basis in economics and is fundamentally concerned with reducing information asymmetry problems between two parties—the signaller and the receiver (Spence, 2002; Connelly et al., 2011; Cotter et al., 2011). The signaller is an insider, such as

executives or managers, who obtain information unavailable to outsiders (Connelly et al., 2011; Albertini, 2019). The receiver is an outsider, or a group of outsiders, who lack information about the organisation but would like to receive this information (Connelly et al., 2011).

Although ST has its roots in the economy and the market, it only includes the investors and the shareholders among the receivers; hence, several studies have subsequently expanded the concept of receivers (Omran & Ramdhony, 2015; Su et al., 2016; Maama and Mkhize, 2020). In particular, in the management and accounting literature, the receivers are generally the stakeholders or stakeholder groups (Connelly et al., 2011; An et al., 2011; Lu and Wang, 2021).

The information asymmetry is reduced when the signaller sends a signal to the receivers. 'A signal can be an observable action, or an observable structure, which is used to indicate the hidden characteristics of the signaller' (An et al., 2011, p. 575). The literature has questioned the kind of signal that the companies send to the market. Stiglitz (2000) highlighted two types of information from which the information asymmetry is derived—information about quality that signals unobservable attributes of the signaller and information about intent that signals behaviour or behavioural intention of the signaller. Starting from this distinction, some studies (Connelly et al., 2011; Albertini, 2019) proposed three different categories of signals: intent, camouflage, and need. Intent signals indicate future actions and refer to the signaller's future strategies. Camouflage signals hide potential signaller's responsibility and are designed to divert the attention of receivers away from a potential vulnerability. Finally, need signals convey some signaller's requirements to the receivers to influence their resource allocation decisions.

Therefore, according to ST, companies should signal their desirable unobservable characteristics and other good qualities to the stakeholders to reduce the information asymmetry between the company itself and its stakeholders and improve companies' transparency (Lu and Wang, 2021). Signalling would lead stakeholders to reassess the company's value and thus make decisions that are more favourable to the company (An et al., 2011).

Companies' unobservable characteristics can be signalled through various mechanisms, such as the voluntary disclosure of positive accounting information (An et al., 2011). In fact, in management and accounting studies, ST has been used to explain why companies voluntarily disclose non-financial information in their annual reports. According to these studies, the voluntary disclosure of non-financial information is driven by the need to preserve or increase the value of the firm, limit the risk of adverse selection (van Zijl et al., 2017), reduce information asymmetry, improve financial performance (Su et al., 2016), improve relationships with the stakeholders (An et al., 2011), and influence the behaviours and actions of stakeholders (Maama and Mkhize, 2020).

Inspired by this insight, companies can use voluntary disclosure of AI information to signal unobservable company characteristics. In particular, given the importance of using AI for value creation, productivity, and sustainability, voluntary disclosure of information about the quality and intent of AI development, deployment, and use can be an effective means for companies to signal their excellence over competitors in terms of technologies, artifacts, and applications, to attract investment. Similarly, the voluntary disclosure of information on the adoption of ethical principles concerning AI can be considered an effective means of signalling to the stakeholders that the company has an ethically superior value and is trustworthy to improve social relations and increase the value of the company among its stakeholders.

Legitimacy theory

Legitimacy theory finds its root in the social and political view of interactions and exchange among actors in the social system (Gray et al., 1995). Corporate legitimacy has been defined as 'a generalized perception or assumption that the actions of an entity are desirable, proper, or appropriate within some socially-constructed system of norms, values, beliefs and definitions' (Suchman, 1995, p. 574). According to LT, a company needs legitimacy in the sense of a social 'licence to operate', to access resources, successfully conduct business, and guarantee the corporation's continued existence (Deegan, 2002). LT suggests that no organisation has an inherent right to exist, but that any business operation is subject to social acceptance (Suchman, 1995). Therefore, companies need to adopt legitimation strategies to guarantee legitimacy, which affects the company's continued existence.

Company disclosure of information, voluntary or mandatory, is one of the means for implementing a legitimacy strategy (Dumay et al., 2015). In fact, several studies (e.g., Campbell et al., 2003; De Villiers and Van Staden, 2006; Bonsón and Flores, 2011; Bonsón and Bednárová, 2013; Hahn and Kühnen, 2013; Solikhah, 2016; Lamboglia et al., 2018; Nielsen and Thomsen, 2018) suggest that LT provides an adequate explanation of non-financial disclosure in most cases; hence, it has become the most widely used explanatory theory in this field.

LT considers the disclosure of information as an attempt to reduce a perceived legitimacy gap to gain, maintain, or restore legitimacy by aligning corporate behaviour with stakeholder expectations. Organisations may gain social acceptance, approval for their acts, and legitimise their actions by engaging in higher transparency (Bonsón and Bednárová, 2013). To obtain legitimacy, companies need to continuously revise their activities in a socially acceptable manner, enhance visibility, and communicate their activities to a broad audience.

Suchman (1995) identifies three types of organisational legitimacy: pragmatic, moral, and cognitive legitimacy. All three types rest on the perception that organisational activities are desirable, adequate, or appropriate within some system of social norms, values, beliefs, and definitions. However, each type of legitimacy entails a different behavioural dynamic. Pragmatic legitimacy is framed as the exchange between an organisation and its audience based on self-interested benefits. Audiences mostly base their assessments on self-regarding utility calculations, and organisations often achieve pragmatic legitimacy by directing tangible rewards to specific constituencies. Moral legitimacy is based on judgements on whether the organisation's activities generate social benefits, and it reflects a positive normative evaluation of the organisation and its activities. Unlike pragmatic legitimacy, moral legitimacy rests on judgements regarding whether the activity is 'the right thing to do' and effectively promotes societal welfare defined by the audience's socially-constructed value system. Finally, cognitive legitimacy is based on cognition rather than interest or evaluation. It is achieved by informing the audience regarding the organisation's practices and objectives and by convincing them, on a conscious and cognitive

level, of their desirability and social acceptability. At this level, legitimacy stems from information and reports based on the mesh of a solid belief system and a consistent real activity.

LT is suited to provide a particular way of interpretation for the communication of information related to AI initiatives towards society and stakeholders. It consents to understand the legitimacy strategy of companies in matching stakeholders' expectations and firms' interests (Dumay et al., 2015). AI information is shared to satisfy the interests of a group of constituents; hence, disclosure may be limited to specific AI projects and/or activities and/or related ethical questions. Pragmatic legitimacy is related to instrumental AI disclosure and, therefore, may be framed as the exchange between an organisation and its audience based on self-interested benefits. Moral legitimacy is value-based and approaches AI disclosure through ethics. In terms of AI disclosure, moral legitimacy requires companies to provide information about the benefits and risks of AI activities from a multi-audience perspective. In doing so, companies gain legitimacy by showing their sensitivity to trustworthy AI. Cognitive legitimacy may produce a taken-for-granted legitimation, and the company 'becomes unassailable by construction' (Suchman, 1995, p. 583) by sharing exhaustive and trustworthy information about AI development, deployment, and use, thus adopting a collective audience perspective.

As one moves from the pragmatic to the moral to the cognitive legitimacy, the achieved legitimacy becomes more difficult to manipulate but also more subtle, profound, and self-sustaining.

The Theoretical Framework

Following the study by An et al. (2011), which combines and integrates several theories into a single theoretical scheme to interpret voluntary intellectual capital disclosure, we have constructed an all-encompassing framework to interpret voluntary AI disclosure practices by organisations. Specifically, starting from the description of VDT, ST and LT, as commented in the previous paragraphs, we identified their common features and the relationship between them. Subsequently, we integrated their concepts to develop a comprehensive framework for explaining AI disclosure, as shown in Fig. 1.

Moving from VDT to ST and LT, the categories of individuals intended as recipients of AI disclosure (who) become progressively wider: from investors to stakeholders, up to society in general.

Furthermore, moving from VDT to ST and LT, companies disclose different types of information (what) considered relevant with respect to their recipients: from disclosing only information about AI initiatives favourable to the organisation to information on unobservable and value-driver characteristics of AI initiatives of the organisation, up to information compliant with AI-related social, legal, and ethical norms.

Finally, passing from VDT to ST and LT, the purpose of the disclosure (why) takes on different configurations: from reporting on AI to pursue specific financial and competitive interests of the organisation to signal the superior quality of the company in developing, deploying, and using AI, to influence the perception of interlocutors in relying on AI trustworthiness.

In addition, by integrating the concepts among the theories and analysing the relationships between them, we have identified some common elements, represented by the bidirectional arrows in Fig. 1. The interrelated concepts of VDT and ST are information asymmetry and signalling. Companies voluntarily disclose information on AI to signal some positive and valuable aspects of their AI initiatives and thus reduce the information asymmetry with investors and, in general, with stakeholders to influence their behaviours and obtain their favour. The interrelated concepts of ST and LT are signalling and organisational legitimacy. Companies voluntarily disclose information about AI to signal their excellence in developing, deploying, and using AI and, meanwhile, try to meet societal

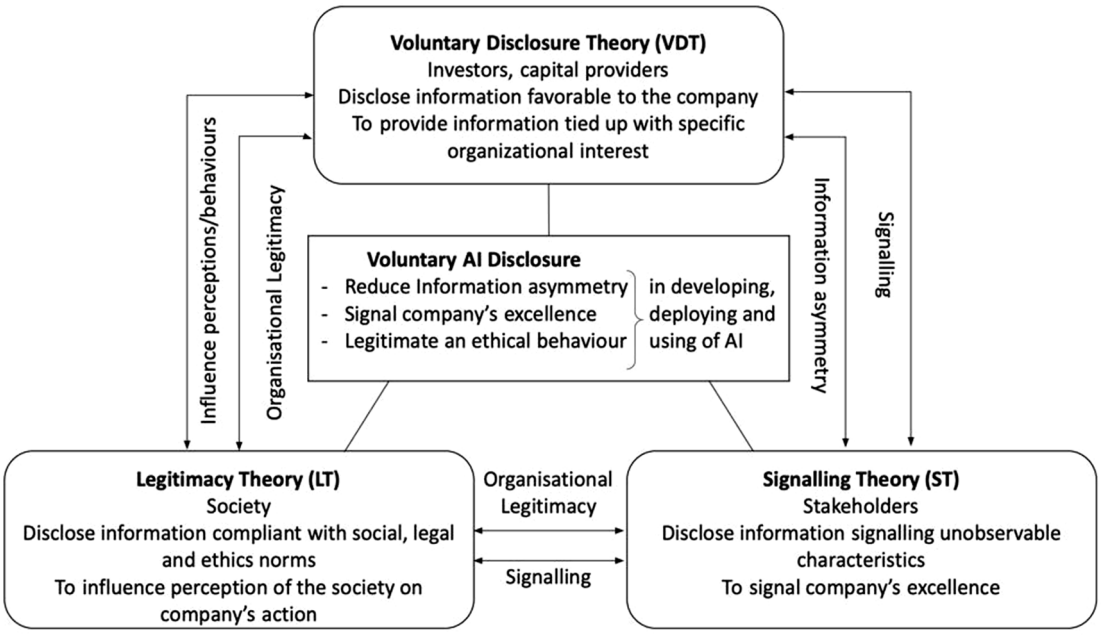


Fig. 1. The integrated theoretical framework on AI disclosure.

expectations and norms to ensure organisational legitimacy through trustworthy AI. Finally, the interrelated concepts of LT and VDT are to influence perceptions/behaviours and organisational legitimacy. Companies decide to disclose AI information favourable to the company to influence or change the behaviour of capital providers and the perceptions of society in general and gain organisational legitimacy.

The main elements of the theories have been integrated into three concepts that can be understood as the rationale behind voluntary AI disclosure: reduce the information asymmetry, signal company's excellence, and legitimate ethical behaviour. Therefore, according to the proposed integrated framework, AI disclosure consists of communicating information about the development, deployment, and use of AI across the company's managerial processes. The objectives are to reduce information asymmetry in the capital market, signal the company's excellence and quality to relevant stakeholders, and legitimate ethical behaviour aligned to social expectations.

4. Research design and methodology

In this section, we describe the study's sample, and the research design based on two phases (Fig. 2): a qualitative-quantitative step based on content analysis and descriptive statistics to address RQ1, RQ2 and RQ3, and a quantitative step based on multiple regression analysis to address RQ4. The combination of these two phases represents an original and valuable methodology to approach AI disclosure.

Sample

The first step of analysis consisted in determining the sample. We chose a representative group of European countries: Germany, Sweden, Finland, France, Spain, and Italy. We addressed the European context, rather than the USA or other countries, for various reasons: European companies follow the same reporting standards for financial and non-financial information, and the European Commission published a set of Ethics Guidelines for Trustworthy AI (European Commission, 2019b), making comparison relatively straightforward. Among the European countries, we chose the first four countries in terms of Gross Domestic Product (GDP) which represent more than 50% (France, Germany, Spain and Italy) (https://ec.europa.eu/eurostat/data/database) and we added the two pioneer countries in the use of AI (Sweden and Finland) (McKinsey & Company, 2019). Within each country, we selected companies listed in the main market indexes (CAC 40, DAX 30, IBEX 35, MIB 40, OMX HELSINKI 25, and OMX STOCKHOLM 30) to assure the availability of official reports and data (e.g., clustering the companies by sectors and company size) and comparability among companies. A total of 200 companies were analysed. The size of the sample is small as the study has an exploratory nature. However, its findings may serve as a starting point for companies interested in AI disclosure.

Content analysis

To examine corporate disclosure regarding AI activities and ethical approaches to AI, we adopt a content analysis methodology, and we analyse the data using descriptive statistics.

Content analysis is a qualitative-quantitative approach for gathering data by extracting useful information from the text and codifying information into predefined categories. It is a systematic and objective method for describing and quantifying phenomena (Krippendorff, 1980; Patton, 2014). Analytical constructs, or rules of inference, are used to move from the text to the proposed research questions (White and Marsh, 2006). This method helps provide new knowledge, insights, representations of facts, and a practical guide to action (Krippendorff, 1980).

This approach seems the most appropriate to analyse the content of the disclosures made by organisations by focusing on what is being reported (Guthrie et al., 2004). Research on CSR typically examines the content of corporate reports for disclosures of social, environmental, and/or ethical issues (Unerman, 2000; Argento et al., 2019; Kumar and Firoz, 2019; Tirado-Valencia et al., 2019).

An essential stage in any content analysis is selecting the relevant documents (Krippendorff, 1980). In the literature, content analysis has been mainly conducted on annual reports as they are an adequate instrument to measure relative positions and trends in reporting (Guthrie et al., 2004) and a useful source of information for communicating and connecting companies with various external and internal stakeholders (Guthrie and Petty, 2000).

We examined the annual reports and/or sustainability reports published on the companies' official websites for the year 2018. These documents were the first to be influenced by the first draft of the EU ethical framework 'Statement on Artificial Intelligence, Robotics and «Autonomous» Systems' (European Group on Ethics in Science and New Technologies, 2018). To enhance the

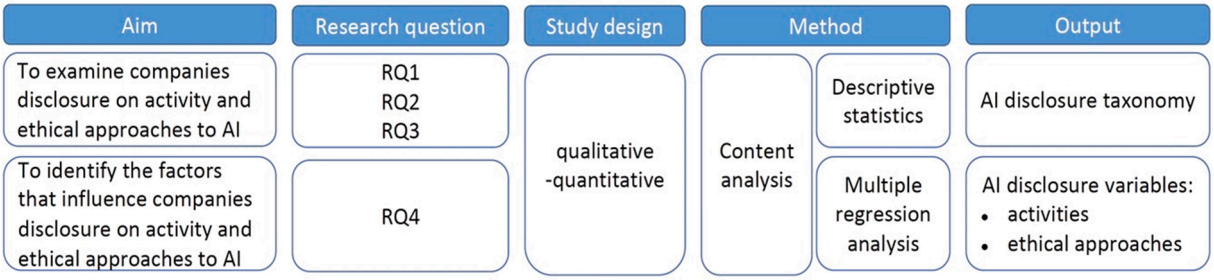


Fig. 2. Research design.

comparability of the data, and due to language constraints, the English version of the reports has been analysed.

All reports were investigated by using specific keywords: 'artificial intelligence', 'machine learning', 'deep learning', and 'big data'. Initially, the keywords have been identified starting from the EU document on AI definition (European Commission, 2019a) and then, they have been revised throughout the analysis.

We considered whole sentences and logical parts of sentences that mention the keywords as a unit of analysis.

The content analysis was conducted manually, with no specific software, and all results were marked and digitally stored.

The process of content analysis started in September 2019 and ended in December 2019.

Before the coding step, we drafted an initial codebook by developing specific categories to analyse the content of the reports. To this end, we classified the mentions to the keywords according to the following categories: general statement, project, unit, product/application, and ethics. These categories are based on our perception of the main topics on which companies report concerning AI. In the first steps of analysis, we employed this provisional scheme; after analysing an initial portion of the sample, no additional categories were introduced. Hence, this taxonomy was validated and employed to develop the empirical analysis of the listed companies.

To cope with subjectivity, an iterative protocol of analysis was established, as content analysis does not proceed linearly and allows flexibility in terms of research design (Harwood and Garry, 2003).

To test the classification scheme and ensure the validity and reliability of the predefined categories, we follow the protocol introduced by Milne and Adler (1999). In particular, the reliability of the coding is a fundamental element in a content analysis as it attributes rigorously to the methodological approach used.

According to Milne and Adler (1999), ensuring reliability in the content analysis involves two issues. First, the content analysis seeks to attest that the coded data set produced from the analysis is reliable. In our analysis, this issue was addressed by using multiple coders and ensuring that any discrepancy among the coders was re-analysed and the differences resolved. The second issue concerns the reliability associated with the coding instruments. This issue was solved by establishing a reliable coding instrument with well-specified decision categories and decision rules. Moreover, well-specified decision categories and rules help to reduce discrepancies between coders.

To test the reliability of the established decision categories, two authors independently coded and analysed an initial portion of the sample. Specifically, the authors analysed the full reports of all companies listed in the IBEX 35. To approve the established classification scheme, the authors compared their findings, identifying some areas of confusion. Subsequently, changes were made to the classification descriptions (Patton, 2014). These changes were then incorporated into the codebook, and the final coding was established. The content was coded according to the following categories (Table 2). The results of this preliminary test were used to create decision rules for the consistent coding of the sample.

Once the codebook and decision categories were validated, three senior-level undergraduate students (research assistants) were involved in the content analysis. Prior to the coding process, the two authors organised preliminary training sessions with research assistants. During these meetings, the authors illustrated the main notions of the research topic, research objectives, use of the coding tool, and an explanation of the different categories that made up the codebook. The authors also detailed the coding process using different reports analysed in the initial sample to highlight different types of information and how to classify the different types of information. At the end of the sessions, each research assistant was given a few reports to attempt coding. After this preliminary coding cycle, the two authors met the assistants and reviewed their coding and categorisation. The authors also addressed specific problems in coding raised by research assistants.

After the initial coding cycle, the research assistants had to report regularly to the two authors via weekly meetings. In these meetings, the research assistants would discuss any problems they encountered with the data. Any ambiguity with the disclosed elements was investigated and resolved. This was necessary to ensure consistency and reliability in the content analysis procedure (Milne and Adler, 1999).

At the end of the analysis process, the research assistants entered the data into a database), which was subsequently reviewed by the two authors to ensure it was error-free.

Variables and measurement

To identify factors affecting companies' reports on their AI activities and/or ethical approaches to AI, we used two dependent variables, 'activity' and 'ethics', as two binary variables. Companies were considered 'active' (activity = 1) if they reported information classified as project, product, or unit. We used the disclosure of these types of information as a proxy for AI activities. Companies were classified as 'non-active' (activity = 0) if they did not report any details on their AI activities or just provided general statements. Companies declaring that they developed or used ethical principles concerning AI in their financial and non-financial reports were considered 'ethical' (ethics = 1, and 0 otherwise).

Table 2
AI disclosure taxonomy.

Category	Description
General statement	Sentences with generic content, referring for example to the context in which the company operates or the spread of new technologies, not providing any specific information regarding AI development, deployment and use.
Project	Specific projects, developed or under development, related to AI
Unit	Specific organisational unit responsible for developing, deploying and managing AI systems employment
Product/application	Specific AI applications (example chatbot, etc.)
Ethics	AI ethical principles

Based on the prior literature (Brammer and Pavelin, 2006; Bonsón and Flores, 2011; Clarkson et al., 2011; Bonsón and Bednárová, 2013; Hahn and Kühnen, 2013; Solikhah, 2016; Maama and Mkhize, 2020), we identified some independent variables that could be potentially associated with reporting on AI activities or ethical approaches to AI, including "industry", "region" and "size".

In terms of 'industry', the sampled companies belong to several sectors. We grouped them into two classes based on their theoretical implications for AI development. We assigned a value equal to 1 to companies operating in the Technology and Telecommunications industries, and 0 otherwise. Concerning the 'region', we grouped the countries into two main areas: Nordic (Finland, Germany, and Sweden) and Southern (France, Italy, and Spain) countries. We assigned a value equal to 1 to companies listed in Nordic stock exchanges, and 0 otherwise.

Finally, to measure the 'company size', we used the natural logarithm of the revenue of each company to mitigate the substantial skewness of this data.

Table 3 summarises all variables and measurements.

5. Findings and results

In this section, we present the main findings regarding AI activities and ethical approaches and the results of the regressions analysis to identify the factors that could influence disclosure in European listed companies. To answer RQ1, an overview of the aggregated disclosures examined is provided, followed by a more detailed analysis of the variables 'activity' and 'ethics' to address RQ2 and RQ3. The results of the statistical tests are used to assess RQ4.

5.1. *AI disclosure (RQ1)*

Table 4 reports the percentage of companies reporting information on AI or general statements.

The highest percentage of companies not reporting on AI is found in Sweden (33.4%), followed by Finland (32.0%), Italy (27.5%), Spain and Germany (20.0%), and France (15.0%). General statements comprise AI information regarding general issues, such as rapid technological change, the impact of new disruptive technologies, the benefits derived from the adoption of new technologies, and the digital transformation, among others. The results indicate that 24.0% of companies in Finland only report on general issues, followed by France, Germany, and Spain (20.0%), while the lowest percentage is found in Sweden (13.3%) and Italy (10.0%). Finally, the share of companies disclosing information on AI is approximately the same in France (65.0%), Italy (62.5%), Spain and Germany (both 60.0%) but lower in Sweden (53.3%) and Finland (44.0%). In the whole sample, 24.0% of companies do not report information on AI, 17.5% only report general statements, while 58.5% declare some kind of AI disclosure.

5.2. *AI activity disclosure (RQ2; RQ4)*

The highest percentage of companies reporting on AI activity is in France (65.0%), followed by Italy (62.5%), Spain and Germany (60.0%), Sweden (53.3%), and Finland (44.0%). In the whole sample, 58.5% of companies report on AI activities.

Table 5 provides a detailed perspective of the reported activities by category.

Specifically, considering the frequency for the whole sample, 34.0% of companies report on AI projects, 43.0% disclose information on AI products, and only 17.5% of companies report about specific AI units. As shown, we also find the same trend among the percentage in the different regions of listing. 'Project' and 'product' are the most frequent categories in the companies' reports. The 'project' that companies most frequently declare are partnerships or AI promotion projects to accelerate digital transformation or specific projects. The 'product' concerns the development of AI applications to solve problems through pattern recognition, prediction, optimisation, recommendation generation, detection, and virtual assistants. The 'unit' category is developed, to some extent, in all countries. The units are autonomous subsidiaries or investee companies and/or laboratories, intended as internal organisational units specialised in AI.

Regarding potential associated factors with AI activity disclosure, the coefficients on all the independent variables are statistically significant (Table 6). Information on AI activity disclosed by European companies is mainly related to revenue. Larger European

Table 3
AI disclosure variables: definition and measurement.

Variables	Name	Description	Score
Dependent	Activity	Company reports information classified as project, product or unit	Dummy variable (if Company is active = 1, if not = 0)
	Ethics	Company declares to develop or use ethical principles on AI in the annual report and/or sustainability report	Dummy variable (if Company develops AI ethical principles = 1; if not = 0)
Independent	Company sector	Company sector as indicated in the Listed Market, grouped according its theoretical implications for AI development in term of technology and telecommunication	Dummy variable (if Company belong to technology or telecommunication sector = 1; if not = 0)
	Region of listing	Regional European area of the Stock Exchange Market in which the company is listed, grouped in northern and southern area.	Dummy variable (if Company is listed in Nordic country = 1; if not = 0)
	Company size	Company size measured by revenue	Natural logarithm of the revenue figures

Table 4
AI disclosure by European listed companies.

	Germany	Finland	Sweden	France	Italy	Spain	Whole sample
No disclosure	20.0%	32.0%	33.4%	15.0%	27.5%	20.0%	24.0%
Just general statement	20.0%	24.0%	13.3%	20.0%	10.0%	20.0%	17.5%
AI disclosure	60.0%	44.0%	53.3%	65.0%	62.5%	60.0%	58.5%

Table 5
AI activity by category.

	Germany	Finland	Sweden	France	Italy	Spain
Projects	40.0%	24.0%	26.7%	32.5%	42.5%	34.3%
Products	46.7%	36.0%	33.3%	50.0%	40.0%	48.6%
Units	23.3%	4.0%	20.0%	22.5%	22.5%	8.6%

Table 6
Factors affecting the AI activity disclosure of European listed companies.

Independent variable	Dependent variable Activity			
	Estimate	Std.Error	z value	Pr (>|z|)
(Intercept)	−0.4041	0.3135	−0.377	0.194717
Sector	1.6634	0.5573	3.053	0.002838 **
Size	0.9536	0.2532	3.795	0.000166 ***
Region	−1.1331	0.3321	−3.22	0.000661 **

** Significant at $p < 0.01$
*** Significant at $p < 0.001$

companies are more likely to report on AI activity. The findings also reveal that companies operating in the Technology and Telecommunications industries and Southern European companies report more information on AI activities.

5.3. AI ethical approaches (RQ3; RQ4)

The highest percentage of companies disclosing AI ethical principles is found in Finland (24.0%), followed by Germany (6.6%), Sweden (3.3%), and Spain (3.3%), while none of the French and Italian companies reports information on ethics. In 2017, the Finnish government introduced an operational programme for turning Finland into a leading country in the application of AI (Steering Group of the Artificial Intelligence Programme, 2017). As a result, companies have been invited to create ethical principles for AI.

Table 7 reports a comparative analysis of ethical pronouncements. For each company, we analysed the disclosed ethical principles and tried to group the most common elements following the categories identified by the AI-HLEG document 'Ethics Guidelines for Trustworthy AI' (European Commission, 2019b): human agency & oversight, technical robustness & safety, privacy & data government, transparency, diversity, non-discrimination & fairness, social & environmental well-being, and accountability. The principle 'continuous reviewing & dialogue' has been added as a result of the proposed content analysis. This addition reflects the need to constantly review AI principles and ways of operating and engage in dialogues with stakeholders for further development of AI activities and AI ethics. Within the nine listed European companies that formally disclose an ethical approach to AI, only Nokia, a Finnish company, declared to fully comply with the EU ethics guidelines.

Regarding potential associated factors with AI ethics disclosure, only two variables are statistically significant: 'industry' and 'region'. Technology and Telecommunications companies and Northern companies are more likely to report if they follow ethical principles concerning AI.

Table 7
AI Ethical approaches.

	Deutsche Telekom	DNA	Kesko	Nokia	Sampo	Sap	Telia	Telefónica	Tieto
Human Agency & Oversight	X	X	X	X	X	X	X	X	X
Technical Robustness & Safety	X	X		X	X	X	X		X
Privacy & Data Governance	X	X	X	X	X	X	*implicit*	X	
Transparency	X	X	X	X	X	X	X	X	X
Diversity, Non-discrimination & Fairness	*implicit*	X	X	X	X	X	X	X	X
Social & Environmental Wellbeing	X		X	X		X		X	X
Accountability	X	*implicit*	X	X	X	*implicit*	X	*implicit*	
Continuous Reviewing & Dialogue			X				X		

Table 8
Factors affecting the AI ethical approaches disclosure of European listed companies.

Independent variable	**Dependent variable Ethics**			
	Estimate	Std.Error	z value	Pr (>|z|)
(Intercept)	−5.8752	1.1664	−5.037	4.73e-07***
Sector	3.8285	0.8562	4.472	7.76e-06***
Size	0.1528	0.5499	0.248	0.7811
Region	1.9568	0.8966	2.183	0.0291*

*Significant at $p < 0.05$
**Significant at $p < 0.01$
***Significant at $p < 0.001$

6. Discussion

This study enhances the understanding of AI activities and ethical approaches by investigating companies that disclose AI information in their official reports. To this end, we first perform a content analysis on the annual and/or sustainability reports on a sample of 200 listed European companies to gain insights into disclosure practices on AI activity and its ethical issues. Subsequently, we perform a descriptive statistical analysis to address RQ1, RQ2, and RQ3. We conduct multiple regression analyses to identify factors influencing companies' disclosure on AI activities and ethical approaches to AI, thus addressing RQ4.

The study's results contribute to the understanding of AI disclosure practices of European listed companies and the underlying motivations in several ways.

The results of the descriptive statistics on AI disclosure (Table 4) show that most European listed companies disclose AI information to some extent in their official reports (RQ1). This reveals their willingness to signal a certain interest in this emerging technology and a quest for a sort of value recognition and organisation legitimacy. This finding also suggests a growing interest in these technologies, although 41.5% of firms are still reporting any information on AI activity or merely general statements. This reporting activity is expected to grow as the spread of AI accelerates.

Furthermore, the content analysis reveals that the amount and type of information disclosed are considerably different (general statements versus AI activity versus ethical issues), also with respect to the capital market examined. The lack of mandatory AI reporting regulation allows companies to decide whether to disclose and the extent and type of information to disclose, depending on their interests and perceptions. As highlighted in the introduction, the adoption of AI could generate many benefits in terms of increased productivity and efficiency; however, it also involves many risks and ethical challenges (European Commission, 2019b). Nevertheless, the analysis found that companies do not disclose any negative information, such as the risks that can arise from adopting AI (e.g. the impact on human rights, privacy, employment, discrimination, among others) (European Commission, 2019b). In this sense, the listed companies intentionally and voluntarily communicate only positive AI information to influence investors' behaviour and reduce information asymmetry (Zhou et al., 2017). Some companies could decide to disclose information about AI activity (Table 5) and ethics (Table 7) to provide more insight on their activities and signal to stakeholders their excellence in developing, deploying, and using AI.

Companies disclose information regarding AI activities—projects, products, and units according to the proposed taxonomy—or general statements to bridge the information asymmetry relating to quality and intent (Stiglitz, 2000). In fact, starting from this distinction and being consistent with the approach of Connelly et al. (2011) and Albertini (2019), the different types of AI information can be interpreted as three categories of signals—intent, camouflage, and need.

The intent signals are composed of information on AI projects, as they signal the company's future actions and AI strategies. Specifically, 34% of European listed companies describe in detail partnerships or AI promotion projects or even specific AI projects, started or in the implementation phase, to signal their intention to accelerate digital transformation and thus show to be more competitive.

General statements can be understood as camouflage signals designed to divert the attention of stakeholders and/or signal involvement in AI issues but hide other details. They include AI information regarding generic issues such as the impact of new

disruptive technologies on businesses in general and the benefits of adopting new technologies. Therefore, 17.5% of companies report only the general statements to signal that they pay attention to issues related to technological developments and the importance of AI. The purpose is to divert the attention of stakeholders from the fact that they have not yet undertaken concrete AI activities or, maybe, they have undertaken them but discretionary and strategically decide to divert stakeholders' attention from their current AI initiatives.

AI product information can be interpreted as need signals, as they refer to requirements that companies send to stakeholders to influence their decisions about resource allocation. To receive more resources, especially financial ones, 43% of companies provide information on investments and AI products intended as the development of specific AI applications, such as virtual assistants and AI recommendation systems. Therefore, companies that disclose information about AI products send a signal to stakeholders, encouraging them to provide their support to the company.

Finally, information about AI units and the adoption of ethical principles concerning AI can be interpreted as signals regarding unobservable attributes of the quality of the company's approach towards AI development, deployment, and use. By providing information on units intended as internal organisational units specialised in AI, 17.5% of the companies provide information on the quality of the structure they have created for AI development, deployment, and use. The information on the adoption of ethical principles concerning AI can be seen as a means of signalling the excellence of a company's approach towards AI over competitors, based on an ethically superior value and trustworthiness framework. However, our results highlight that the adoption of ethical approaches towards AI is not widespread as only 4.5% of companies report ethical principles.

Additionally, the results of the AI activity analysis (Table 5) and AI ethical approaches (Table 7) show that most listed companies examined use AI (RQ2) as a relevant technology to develop products, projects, new organisational units and, in some cases, design internal ethical procedures and external dialogues with stakeholders (RQ3). Owing to the relevance of AI systems and the risks and benefits they could generate (European Commission, 2018), most companies respond promptly to the EU call for communicating information to stakeholders about the AI systems. They disclose information about AI activities and ethics as a strategic means to obtain social acceptance and legitimation, preserving the company's interests. Matching our results with three different types of organisational legitimacy (Suchman, 1995), we find that most listed companies adopt a pragmatic legitimacy by disclosing information about certain projects, and/or products, and/or units to engage a specific audience and interests. Only nine companies adopt a moral legitimacy approach, disclosing some information about the ethical principles of their involvement in AI activities to obtain a multi-audience legitimation based on the general social benefits of their AI activities. None of the sampled companies adopts the cognitive legitimacy approach. At this level, trustworthy AI disclosure may be defined as a communicative practice of companies to gain legitimacy, meshing information on AI activities and ethical approaches, into their business operations and core strategies, in close collaboration with their audiences.

The results of the regression analysis explain the factors associated with the reporting on AI activity and ethical approaches towards AI (RQ4). All the independent variables have a significant impact on AI activity disclosure (Table 6). The largest companies present a higher level of reporting, are more visible, and therefore, are exposed to numerous stakeholders. Hence, the disclosure of AI information can be regarded as an instrument to ensure legitimacy. In addition, larger companies invest sufficient financial resources in the automation of processes. Therefore, the voluntary disclosure of information on AI technologies, applications, and projects can be seen as a means of signalling positive information to attract capital providers and thus reducing the information asymmetry with shareholders. Companies in the Technology and Telecommunications industries are reporting more on AI activities, as expected, as these industries present a high degree of technology adoption, thus developing or intensively using AI. Furthermore, companies belonging to these industries have a particular interest in reporting information on their AI activities because the receivers—the company's stakeholders—are more sensitive and pay more attention to these types of information. Finally, Southern countries' companies are more likely to disclose information on AI activity. Companies in these countries are likely to be subject to greater pressure from investors and stakeholders, therefore they disclose more information to meet the information needs that derive from the traditions of their stakeholders (Hahn and Kühnen, 2013)

Regarding ethical approaches, the regression analysis indicates a significant association of region and industry with ethical disclosure (Table 8). Most companies reporting ethical principles are in the Northern region and belong to the Technology and Telecommunications industries. Most companies (66%) disclosing ethical principles concerning AI are Finnish, probably due to the abovementioned governmental initiative (Steering Group of the Artificial Intelligence Programme, 2017), which seems to have influenced the development of ethical guidelines concerning AI. Regarding the industry, 77% of the companies developing AI ethical approaches belong to the Technology and Telecommunications industries. These industries are highly sensitive to AI problems; hence, they typically face higher pressure from stakeholders to develop, deploy, and use AI principles. In this context, for example, SAP and Nokia are directly engaged with EU-HLEG, as their companies' members belong to the expert group, while Telefonica and Deutsche Telekom seem to be leaders in the field.

7. Conclusions, limitations and future research

This study answers the call for further investigation of AI activities and their ethical implications. It focuses on the identification and clarification of the concepts and fields involved in AI activities, the empirical analysis and real use of AI, and the related ethical issues. It also investigates how theoretical AI frameworks may be translated into 'practical implementation of ethics into AI systems' (Vakkuri and Abrahamsson, 2018, p. 5).

We provide evidence of AI disclosure. The present study makes several contributions to the AI literature. To the best of our knowledge, this is the first empirical study exploring the AI disclosure practices of leading European companies and is one of the few studies on AI disclosure, as a type of non-financial disclosure which has not been explored yet in the literature. On the one hand, the

study reviews the state of the art in terms of the AI projects that companies launch, the labs or units they create, and the products they develop or use. On the other hand, it explores the AI ethical principles that companies adopt, if any, to guarantee that algorithms and other AI tools are used fairly regarding privacy, human rights, transparency, lack of bias, and accountability, among others.

Hence, this study's results may serve as a starting point for researchers and companies interested in AI. Unlike existing studies, we propose a first definition of the topic and a taxonomy that can be used in further research on AI disclosure and can contribute to the development of KPIs in the field.

Furthermore, this study provides a theoretical framework integrating some traditional theories, such as VDT, ST, and LT, specifically drawn to interpret AI disclosure practices, which can help with a further in-depth exploration of AI disclosure combining concurrent perspectives. The proposed integrated model is a promising framework that can be used to explain and predict AI disclosure practices by listed companies.

Several limitations in this research must be acknowledged, regarding the empirical evidence and the theoretical framework. First, we only considered the disclosure of annual and sustainability reports. Future studies may also analyse companies' websites (the development of AI is possibly higher than what has been reported). Second, this is an exploratory study; thus, the analysis needs to be extended to a longer period to assess the impact of the EU ethics guidelines (the draft has been published in 2018, but the official version dates back to 2019). Finally, the size of the sample limits the generalizability of the study's results. Our data collection is limited to European companies; therefore, future studies may extend the proposed approach and apply it to a global context, improving the generalizability of the results.

Regarding the theoretical framework, it considers only three theories and can be enlarged to include some other theoretical perspectives that are relevant for AI disclosure. Furthermore, we consider the who, what, and why of AI disclosure, ignoring other relevant aspects such as the cost of disclosure.

Future research should address the development of performance indicators for AI (by sector) to assess the use of this technology. Best practices should be identified, and benchmarking should be implemented. Different practices should be unified, and companies should be asked about the implications of the software they are using, their fair use, and what they are doing to control it. Future studies should also map the evolution of the extent of AI disclosure practices by European companies and/or expand the analysis to all European countries or other contexts, such as the USA, UK, and China. The proposed categories may be extensively used as they adequately represent the main topics used by companies to report about AI.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

References

Albertini, E., 2019. Integrated reporting: an exploratory study of French companies. J. Manag. Gov. 23 (2), 513–535. https://doi.org/10.1007/s10997-018-9428-6.

Allen, C., Smit, I., Wallach, W., 2005. Artificial morality: Topdown, bottom-up, and hybrid approaches. Ethics Inf. Technol. 7 (3), 149–155. https://doi.org/10.1007/s10676-006-0004-4.

An, Y., Davey, H., Eggleton, I.R., 2011. Towards a comprehensive theoretical framework for voluntary IC disclosure. J. Intell. Capit. 12 (4), 571–585. https://doi.org/10.1108/14691931111181733.

Annoni, A., Benczur, P., Bertoldi, P., Delipetrev, B., De Prato, G., Feijoo, C., Fernandez Macias, E., Gomez Gutierrez, E., Iglesias Portela, M., Junklewitz, H., Lopez Cobo, M., Martens, B., Figueiredo Do Nascimento, S., Nativi, S., Polvora, A., Sanchez Martin, J.I., Tolan, S., Tuomi, I. and Vesnic Alujevic, L., 2018. Artificial Intelligence: A European Perspective. Craglia, M. editor(s). doi:10.2760/11251.

Argento, D., Grossi, G., Persson, K., Vingren, T., 2019. Sustainability disclosures of hybrid organizations: Swedish state-owned enterprises. Medit. Accountancy Res. 27 (4), 505–533. https://doi.org/10.1108/MEDAR-07-2018-0362.

Bonsón, E., Bednárová, M., 2013. Corporate LinkedIn practices of Eurozone companies. Online Information Rev. 37 (6), 969–984. https://doi.org/10.1108/OIR-09-2012-0159.

Bonsón, E., Flores, F., 2011. Social media and corporate dialogue: the response of global financial institutions. Online Information Rev. 35 (1), 34–49. https://doi.org/10.1108/14684521111113579.

Brammer, S., Pavelin, S., 2006. Voluntary environmental disclosures by large UK companies. J. Bus. Finance Account. 33 (7–8), 1168–1188. https://doi.org/10.1111/j.1468-5957.2006.00598.x.

Brey, P.A., 2012. Anticipating ethical issues in emerging IT. Ethics Inf. Technol. 14 (4), 305–317. https://doi.org/10.1007/s10676-012-9293-y.

Campbell, D., Craven, B., Shrives, P., 2003. Voluntary social reporting in three FTSE sectors: a comment on perception and legitimacy. Account., Audit. Accountability J. 16 (4), 558–581. https://doi.org/10.1108/09513570310492308.

Cho, C.H., Laine, M., Roberts, R.W., Rodrigue, M., 2015. Organized hypocrisy, organizational façades, and sustainability reporting. Accounting. Organ. Soc. 40, 78–94. https://doi.org/10.1016/j.aos.2014.12.003.

Clarkson, P.M., Overell, M.B., Chapple, L., 2011. Environmental reporting and its relation to corporate environmental performance. Abacus 47 (1), 27–60. https://doi.org/10.1111/j.1467-6281.2011.00330.x.

Connelly, B.L., Certo, S.T., Ireland, R.D., Reutzel, C.R., 2011. Signaling theory: A review and assessment. J. Manag. 37 (1), 39–67. https://doi.org/10.1177/0149206310388419.

Cotter, J., Lokman, N., Najah, M.M., 2011. Voluntary disclosure research: which theory is relevant? J. Theoretical Account. Res. 6 (2), 77–95.

De Villiers, C., Van Staden, C.J., 2006. Can less environmental disclosure have a legitimising effect? Evidence from Africa. Accounting. Organ. Soc. 31 (8), 763–781. https://doi.org/10.1016/j.aos.2006.03.001.

Deegan, C., 2002. Introduction: the legitimising effect of social and environmental disclosures- a theoretical foundation. Account., Audit. Accountability J. 15 (3), 282–311. https://doi.org/10.1108/09513570210435852.

Dumay, J., Frost, G., Beck, C., 2015. Material legitimacy. J. Account. Org. Change. 11 (1), 2–23. https://doi.org/10.1108/JAOC-06-2013-0057.

Dye, R.A., 2001. An evaluation of "essays on disclosure" and the disclosure literature in accounting. J. Account. Econ. 32 (1–3), 181–235. https://doi.org/10.1016/S0165-4101(01)00024-6.

European Commission, 2018. Artificial Intelligence for Europe. COM 237. https://ec.europa.eu/transparency/regdoc/rep/1/2018/EN/COM-2018-237-F1-EN-MAIN-PART-1.PDF/.

European Commission, 2019a. A Definition of AI: Main Capabilities and Disciplines. https://ec.europa.eu/digital-single-market/en/news/definition-artificial-intelligence-main-capabilities-and-scientific-disciplines/.
European Commission, 2019b. Ethics Guidelines for Trustworthy AI. https://ec.europa.eu/digital-single-market/en/news/ethics-guidelines-trustworthy-ai/.
European Group on Ethics in Science and New Technologies, 2018. Statement on Artificial Intelligence, Robotics and "Autonomous" Systems. http://ec.europa.eu/research/ege/pdf/ege_ai_statement_2018.pdf/.
Gray, R., Kouhy, R., Lavers, S., 1995. Corporate social and environmental reporting: a review of the literature and a longitudinal study of UK disclosure. Account., Audit. Accountability J. 8 (2), 47–77. https://doi.org/10.1108/09513579510146996.
Guthrie, J., Petty, R., 2000. Are companies thinking smart? Australian CPA 70 (6), 62–65.
Guthrie, J., Petty, R., Yongvanich, K., Ricceri, F., 2004. Using content analysis as a research method to inquire into intellectual capital reporting. J. Intell. Capit. 5 (2), 282–293. https://doi.org/10.1108/14691930410533704.
Hahn, R., Kühnen, M., 2013. Determinants of sustainability reporting: a review of results, trends, theory, and opportunities in an expanding field of research. J. Clean. Prod. 59, 5–21. https://doi.org/10.1016/j.jclepro.2013.07.005.
Harwood, T.G., Garry, T., 2003. An overview of content analysis. Mark. Rev. 3 (4), 479–498. https://doi.org/10.1362/146934703771910080.
Júnior, J.F.A., Oliveira, M.C., Ponte, V.M.R., de Sousa Ribeiro, M., 2014. Social disclosure of Brazilian and UK firms in light of stakeholder theory, legitimacy theory and voluntary disclosure theory. Advanc. Scient. Appl. Account. 7 (2), 175–200. https://doi.org/10.14392/asaa.2014070201.
Krippendorff, K., 1980. Content Analysis: An Introduction to Its Methodology. Sage, London.
Kumar, P., Firoz, M., 2019. Accounting for certified emission reductions (CERs) in India: An analysis of the disclosure and reporting practices within the financial statements. Medit. Accountancy Res. 28 (2), 365–389. https://doi.org/10.1108/MEDAR-01-2019-0428.
Lamboglia, R., Fiorentino, R., Mancini, D., Garzella, S., 2018. From a garbage crisis to sustainability strategies: The case study of Naples' waste collection firm. J. Clean. Prod. 186, 726–735. https://doi.org/10.1016/j.jclepro.2018.03.151.
Lu, J., Wang, J., 2021. Corporate governance, law, culture, environmental performance and CSR disclosure: A global perspective. J. Int. Financial Markets., Inst. Money. 70, 101264 https://doi.org/10.1016/j.intfin.2020.101264.
Maama, H., Mkhize, M., 2020. Integration of Non-Financial Information into Corporate Reporting: A Theoretical Perspective. Acad. Account. Financial Studies J. 24 (2), 1–15.
Martin, K., 2019. Ethical Implications and Accountability of Algorithms. J. Bus. Ethics. 160, 835–850. https://doi.org/10.1007/s10551-018-3921-3.
McKinsey & Company, 2019. Notes from the AI frontier. Trackling Europe's gap in digital and AI. https://www.mckinsey.com/~/media/mckinsey/featured%20insights/artificial%20intelligence/tackling%20europes%20gap%20in%20digital%20and%20ai/mgi-tackling-europes-gap-in-digital-and-ai-feb-2019-vf.pdf/.
Milne, M.J., Adler, R.W., 1999. Exploring the reliability of social and environmental disclosures content analysis. Account., Audit. Accountability J. 12 (2), 237–256. https://doi.org/10.1108/09513579910270138.
Munoko, I., Brown-Liburd, H.L., Vasarhelyi, M., 2020. The ethical implications of using artificial intelligence in auditing. J. Bus. Ethics. 167, 209–234. https://doi.org/10.1007/s10551-019-04407-1.
Nielsen, A.E., Thomsen, C., 2018. Reviewing corporate social responsibility communication: a legitimacy perspective. Corp. Commun.: Int. J. 23 (4), 492–511. https://doi.org/10.1108/CCIJ-04-2018-0042.
Omran, M. A., Ramdhony, D., 2015. Theoretical perspectives on corporate social responsibility disclosure: a critical review. Int. J. Account. Financial Report. 5 (2), 38-55. 10.5296/ ijafr.v5i2.8035.
Palm, E., Hansson, S.O., 2006. The case for ethical technology assessment (eTA). Technol. Forecast. Soc. Chang. 73, 543–558. https://doi.org/10.1016/j.techfore.2005.06.002.
Patton, M.Q., 2014. Qualitative research & evaluation methods: Integrating theory and practice. Sage publications.
Pwc, 2017. What's the real value of AI for your business and how can you capitalise?. https://www.pwc.com/gx/en/issues/analytics/assets/pwc-ai-analysis-sizing-the-prize-report.pdf/.
Rotolo, D., Hicks, D., Martin, B.R., 2015. What is an emerging technology? Res. Policy 44 (10), 1827–1843. https://doi.org/10.1016/j.respol.2015.06.006.
Shehata, N.F., 2014. Theories and determinants of voluntary disclosure. Account. Finance Res. (AFR) 3 (1), 18–26.
Solikhah, B., 2016. An Overview of Legitimacy Theory on the Influence of Company Size and Industry Sensitivity towards CSR Disclosure. Int. J. Applied Bus. Econ. Res. 14 (5), 3013–3023.
Spence, M., 2002. Signaling in retrospect and the informational structure of markets. American Econ. Rev. 92 (3), 434–459. https://doi.org/10.1257/00028280260136200.
Stahl, B.C., Heersmink, R., Goujon, P., Flick, C., van den Hoven, J., Wakunuma, K., Ikonen, V., Rader, M., 2010. Identifying the ethics of emerging information and communication technologies: An essay on issues, concepts and method. Int. J. Technoethics 1 (4), 20–38. https://doi.org/10.4018/jte.2010100102.
Steering Group of the Artificial Intelligence Programme, 2017. Finland's age of artificial intelligence. Turning Finland into a leading country in the application of artificial intelligence. Objective and recommendations for measures. https://julkaisut.valtioneuvosto.fi/bitstream/handle/10024/160391/TEMrap_47_2017_verkkojulkaisu.pdf/.
Stiglitz, J.E., 2000. The contributions of the economics of information to twentieth century economics. Quart. J. Econ. 115 (4), 1441–1478. https://doi.org/10.1162/003355300555015.
Su, W., Peng, M.W., Tan, W., Cheung, Y.L., 2016. The signaling effect of corporate social responsibility in emerging economies. J. Bus. Ethics. 134 (3), 479–491. https://doi.org/10.1007/s10551-014-2404-4.
Suchman, M.C., 1995. Managing legitimacy: Strategic and institutional approaches. Acad. Manag. Rev. 20 (3), 571–610. https://doi.org/10.5465/amr.1995.9508080331.
Tirado-Valencia, P., Cordobés-Madueño, M., Ruiz-Lozano, M., De Vicente-Lama, M., 2019. Integrated thinking in the reporting of public sector enterprises: a proposal of contents. Medit. Accountancy Res. 28 (3), 435–453. https://doi.org/10.1108/MEDAR-03-2019-0458.
Unerman, J., 2000. Methodological issues - Reflections on quantification in corporate social reporting content analysis. Account., Audit. Accountability J. 13 (5), 667–681. https://doi.org/10.1108/09513570010353756.
United Nations, 2015. Transforming our World: the 2030 Agenda for Sustainable Development. https://sustainabledevelopment.un.org/content/documents/21252030%20Agenda%20for%20Sustainable%20Development%20web.pdf/.
Vakkuri, V., Abrahamsson, P., 2018. The key concept of ethics of artificial intelligence. A keywords based systematic mapping study. In: In 2018 IEEE International Conference on Engineering, Technology and Innovation (ICE/ITMC), pp. 1–6. https://doi.org/10.1109/ICE.2018.8436265.
Vakkuri, V., Kemell, K. K., & Abrahamsson, P., 2019b. AI ethics in industry: a research framework. arXiv preprint arXiv:1910.12695.
van Zijl, W., Wöstmann, C., Maroun, W., 2017. Strategy disclosures by listed financial services companies: Signalling theory, legitimacy theory and South African integrated reporting practices. South Afr. J Bus. Manag. 48 (3), 73–85. https://doi.org/10.4102/sajbm.v48i3.37.
Verrecchia, R.E., 1983. Discretionary disclosure. J. Account. Econ. 5, 179–194. https://doi.org/10.1016/0165-4101(83)90011-3.
Verrecchia, R.E., 2001. Essays on disclosure. J. Account. Econ. 32 (1–3), 97–180. https://doi.org/10.1016/S0165-4101(01)00025-8.
White, M.D., Marsh, E.E., 2006. Content analysis: A flexible methodology. Library Trends 55 (1), 22–45. https://doi.org/10.1353/lib.2006.0053.
Wieringa, M., 2020. What to account for when accounting for algorithms. A systematic literature review on algorithmic accountability. FAT* '20: Proceedings of the 2020 Conference on Fairness, Accountability, and Transparency. 1–18. 10.1145/3351095.3372833.
Wright, S.A., Schultz, A.E., 2018. The rising tide of artificial intelligence and business automation: developing an ethical framework. Bus. Horiz. 61 (6), 823–832. https://doi.org/10.1016/j.bushor.2018.07.001.
Wright, D., 2011. A framework for the ethical impact assessment of information technology. Ethics Info. Technol. 13 (3), 199–226. https://doi.org/10.1007/s10676-010-9242-6.
Zhou, S., Simnett, R., Green, W., 2017. Does integrated reporting matter to the capital market? Abacus 53 (1), 94–132. https://doi.org/10.1111/abac.12104.